

Maths

Unit 6 - Fractions and Percentages.

WALT:

6Nf.02: Understand that proper and improper fractions can act as operators.

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I can define the terms proper fraction, improper fraction and operator.



I can use a fraction as an operator to find amounts.



I can use a fraction as an operator to compare amounts.

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Retrieval Practice

Last Lesson

There are 8 pizzas shared between 5 people. What calculation represents this?

Last Week

Ms. Colette says, "The difference between -5 and -9 is -4." Is she correct? Explain your reasoning.

Last Unit

What do the letters represent?

- a) $3 - a = -2$
- b) $f + 12 = 30$
- c) $6a = 24$

Last Year

Here are ten shapes.



What proportion of the shapes are circles?

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Starter – Define these terms to your partner:

Proper fraction A fraction in which the numerator (top number) is greater than the denominator (bottom number.)

Improper fraction A fraction in which the denominator (bottom number) is greater than the numerator (top number.)

Operator A command that tells you what to do with the numbers. E.g. $+$ $-$ $\times$ $\div$

WALT: Understand that proper and improper fractions can act as operators.

Think about what we learned last lesson.

What operator does a fraction act as?



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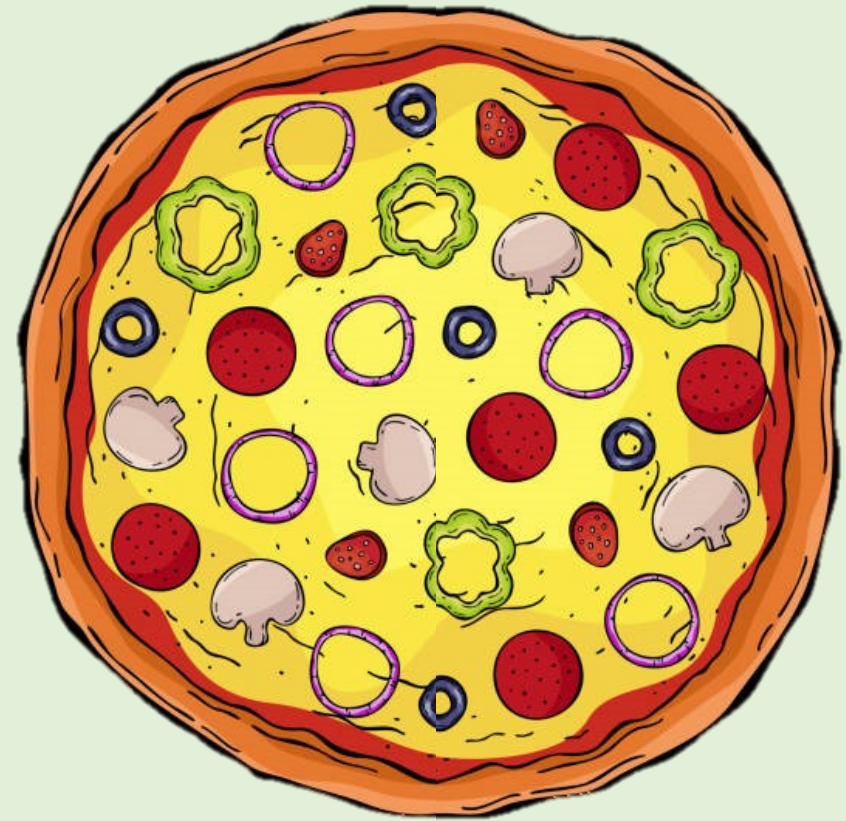
If Mr. Jon eats $\frac{1}{2}$ of a pizza that's **one** pizza divided into **two** equal parts.

1 / **2**

One pizza

Divided

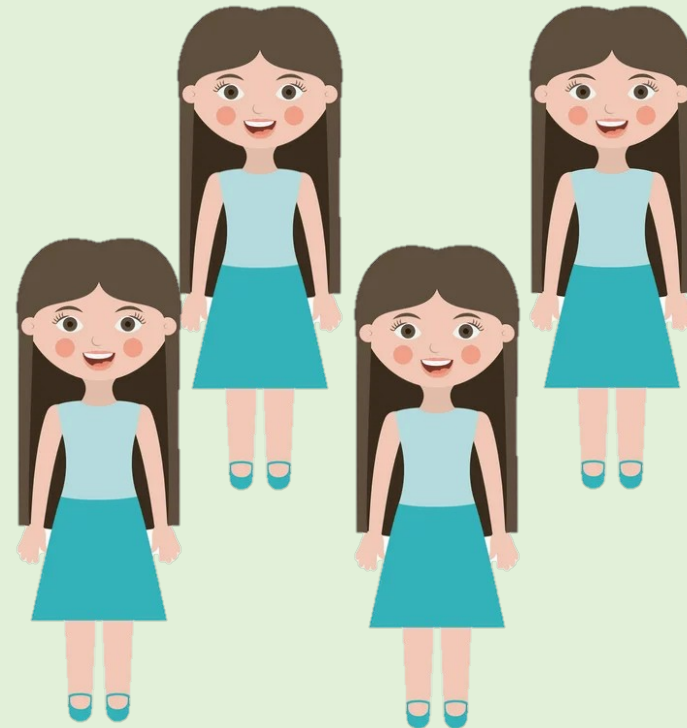
Into two equal parts



WALT: Understand that proper and improper fractions can act as operators.

We can also use improper fractions as operators.

If I have 5 cakes divided by 4 people how much cake does each person get?



$$\frac{5}{4}$$

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If 7 people share 9 cakes, how much would they get each?

$$9 \div 7 = \frac{9}{7}$$

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We can use fractions as operators to find amounts.

What is $\frac{3}{4}$ of 12?

First we divide by the denominator.

$$12 \div 4 = 3$$

Then we multiply by the numerator.

$$3 \times 3 = 9$$

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Check your understanding

1. 6 children share 5 pizzas equally. How much pizza does each child get?
2. What is $\frac{4}{7}$ of 28?
3. Would you rather be given $\frac{3}{4}$ of \$80 or $\frac{9}{5}$ \$30?
Why?

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Check your understanding

1. 6 children share 5 pizzas equally. How much pizza does each child get?

$$\frac{5}{6}$$

2. What is $\frac{4}{7}$ of 28?

$$16$$

3. Would you rather be given $\frac{3}{4}$ of \$80 or $\frac{9}{5}$ \$30?
Why?

$$\frac{3}{4} \text{ of } \$80 = \$60$$

$$\frac{9}{5} \text{ of } \$30 = \$54$$

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1. Use the terms "proper fraction", "improper fraction" and "operator" to complete these sentences.

a) When we calculate $\frac{3}{4}$ we are using the fraction as an _____.

b) A _____ is when the numerator is smaller than the denominator.

c) $\frac{7}{5}$ is an example of a _____.



2. Calculate:

a) $\frac{1}{2}$ of 16

b) $\frac{3}{4}$ of 48

c) $\frac{3}{10}$ of 250

d) $\frac{5}{2}$ of 34

e) $\frac{9}{5}$ of 35

f) $\frac{11}{9}$ of 72



3. a) Mr. Jon wanted to ride his motorcycle 400km in 1 day. He managed to ride $\frac{7}{8}$ of this amount. Ms. Colette wanted to ride her motorcycle 300km in 1 day. She managed to ride $\frac{12}{10}$ of this amount. Who rode their motorcycle further? Explain how you know.

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1. Use the terms "proper fraction", "improper fraction" and "operator" to complete these sentences.

a) When we calculate $\frac{3}{4}$ we are using the fraction as an operator.
proper

b) A fraction is when the numerator is smaller than the denominator.

c) $\frac{7}{5}$ is an example of a improper fraction.

WALT: Understand that proper and improper fractions can act as operators.



2. Calculate:

a) $\frac{1}{2}$ of 16 **8**

b) $\frac{3}{4}$ of 48 **36**

c) $\frac{3}{10}$ of 250 **75**

d) $\frac{5}{2}$ of 34 **85**

e) $\frac{9}{5}$ of 35 **63**

f) $\frac{11}{9}$ of 72 **88**

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3. a) Mr. Jon wanted to ride his motorcycle 400km in 1 day. He managed to ride $\frac{7}{8}$ of this amount. Ms. Colette wanted to ride her motorcycle 300km in 1 day. She managed to ride $\frac{12}{10}$ of this amount. Who rode their motorcycle further? Explain how you know.

Ms. Colette. 400km divided by 8 is 50km. 50km multiplied by 7 is 350km so Mr. Jon rode 350km. 300km divided by 10 is 30km. 30km multiplied by 12 is 360km so Ms. Colette rode 360km. 360km > 350km so Ms. Colette rode further.

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